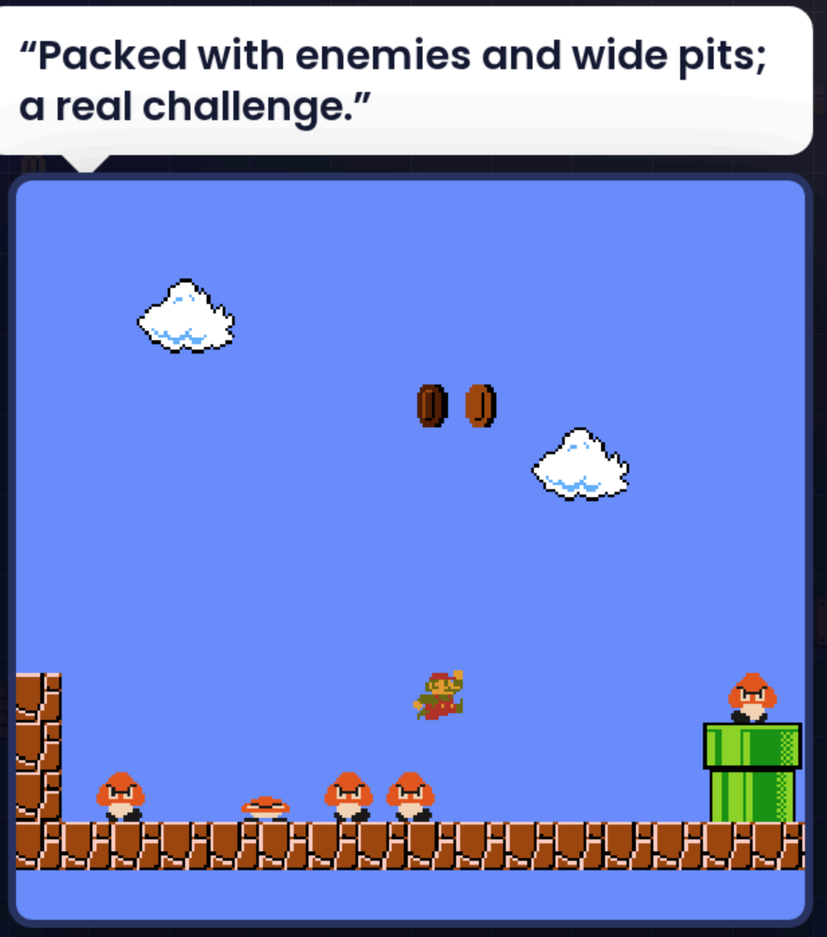
POSTER SESSION · PROCEDURAL CONTENT GENERATION

Did You Get the Level You Asked For?

Bridging LLMs and PCGRL for *Super Mario Bros.* level generation

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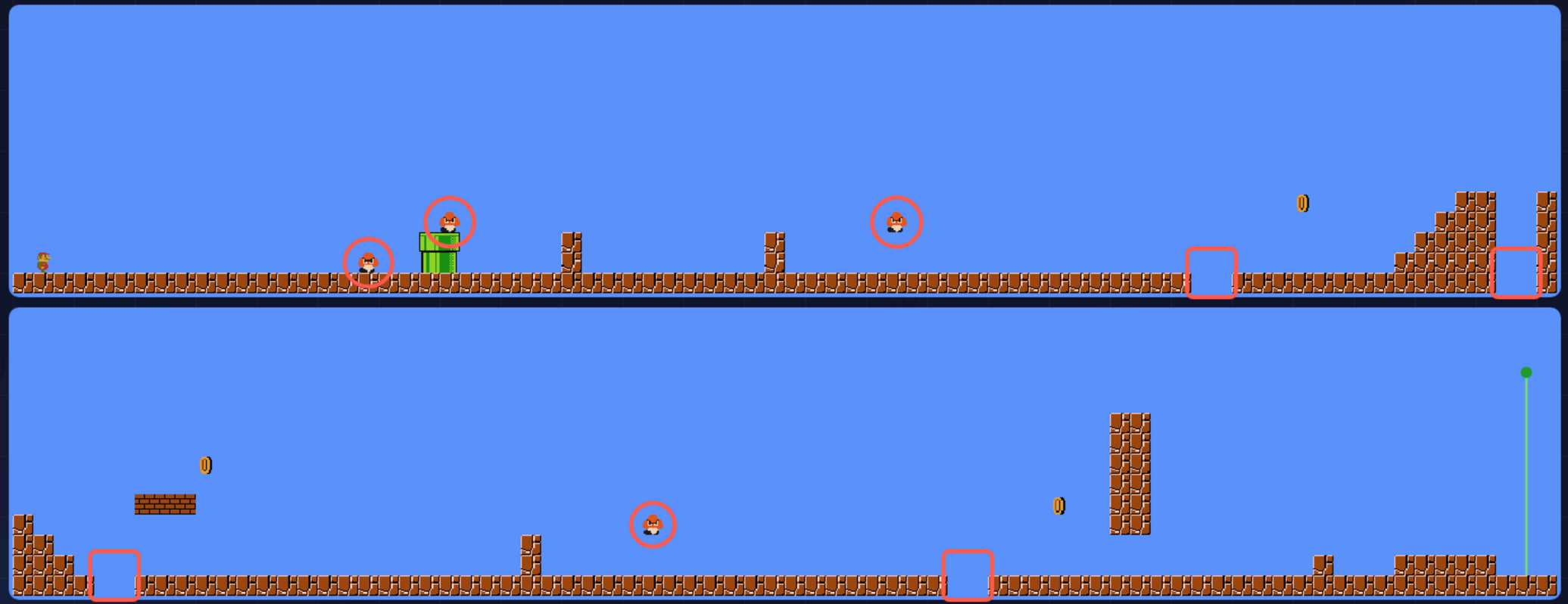


BUILT FROM THE REQUEST ABOVE · A* AGENT PLAYING

A DESIGNER ASKS

“Make me a flat, empty, smooth level.”

...and a corpus-based generator hands back:



4 enemies

4 gaps

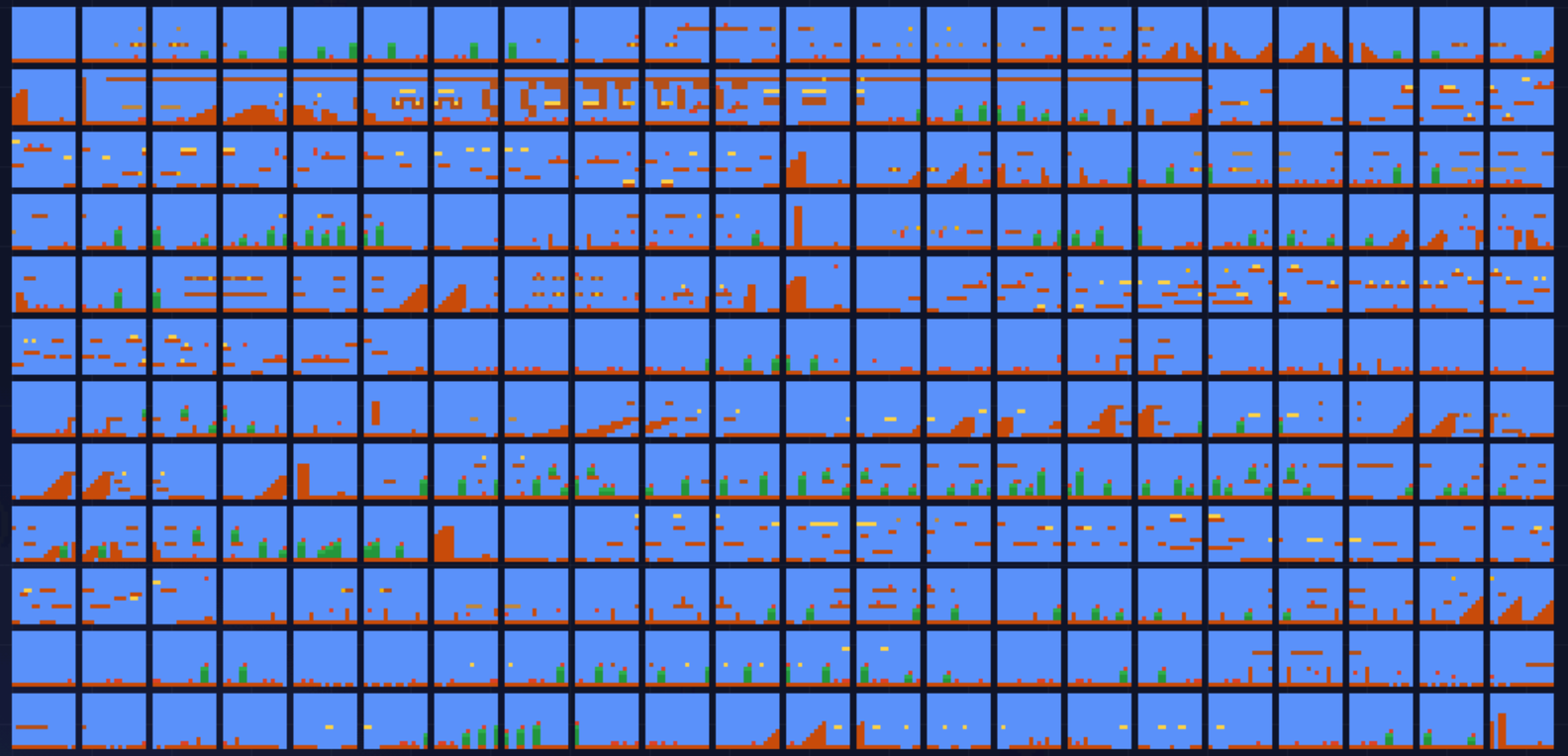
3 coins

3 blocks

Level stitched only from corpus chunks by our policy, before any edit (target: every count 0, smoothness 1).

WHY IT HAPPENS

Generators can only recombine what the corpus has.



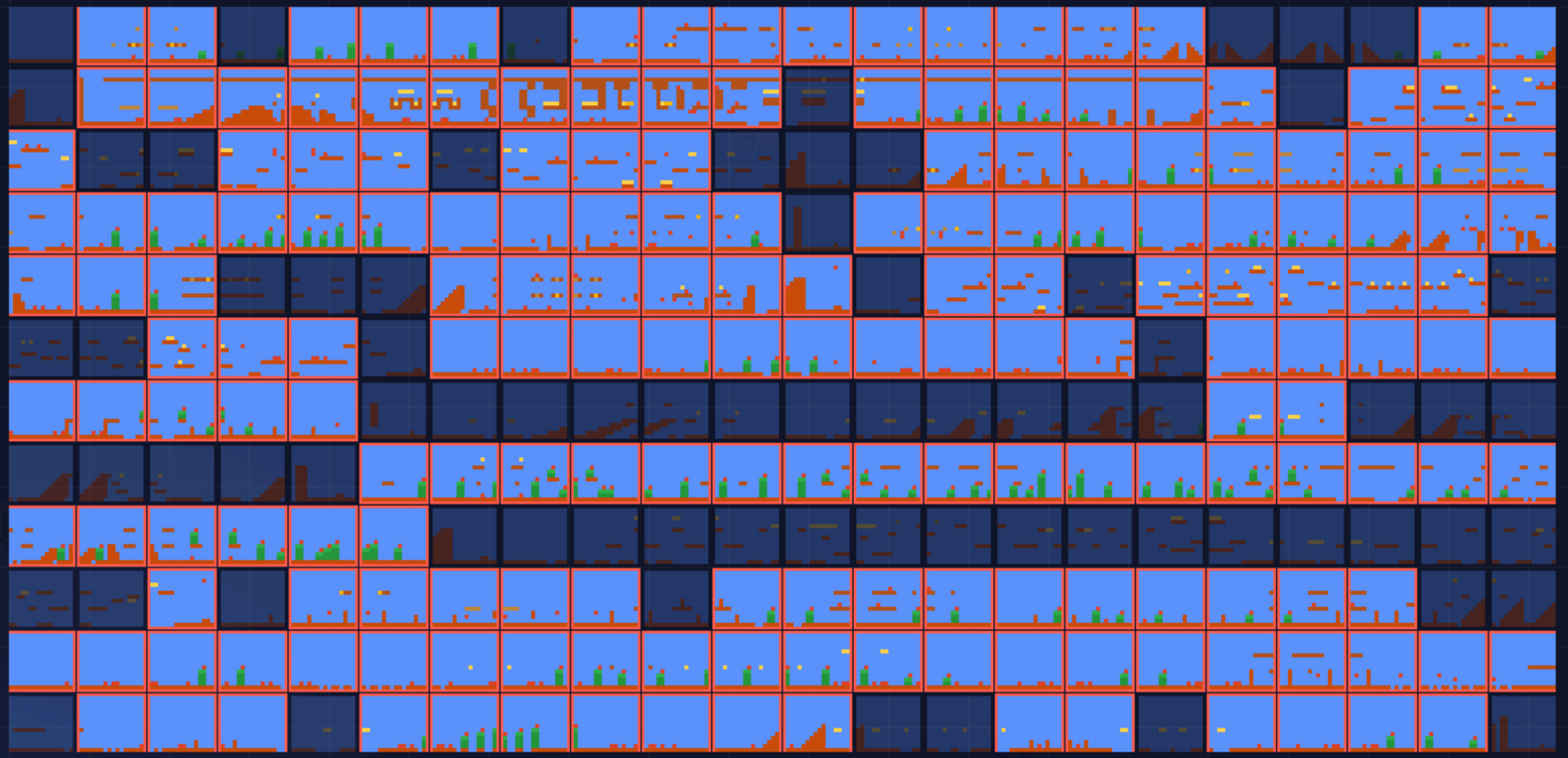
264

corpus chunks, 16 columns each, cut from 15 *Super Mario Bros.* levels

Annotated chunk library: VGLC overworld levels, stride 8 (12 start, 241 middle, 11 ending).

WHY IT HAPPENS

Generators can only recombine what the corpus has.



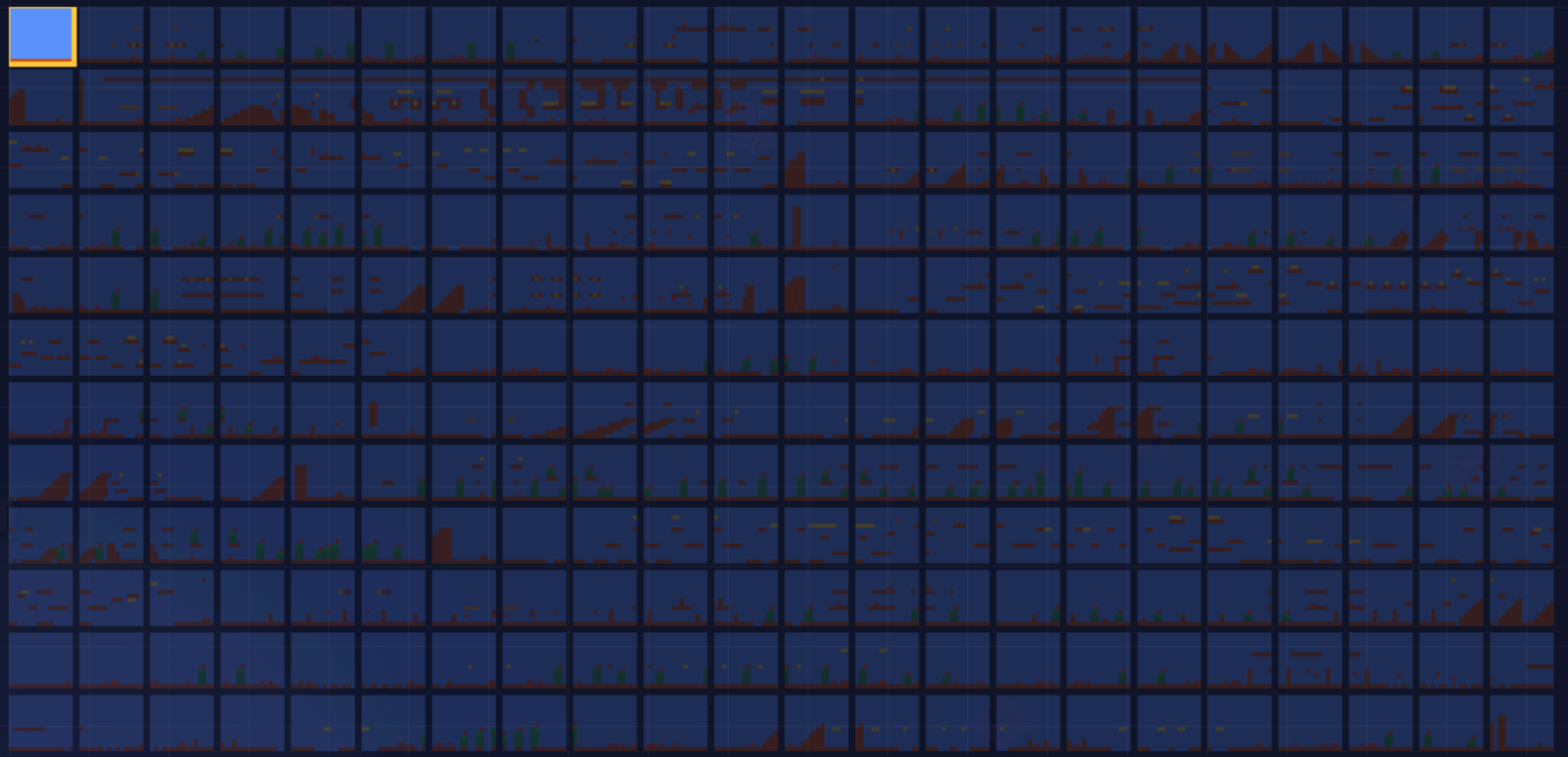
190

of them contain an enemy (72%)

Annotated chunk library: VGLC overworld levels, stride 8 (12 start, 241 middle, 11 ending).

WHY IT HAPPENS

Generators can only recombine what the corpus has.



0 of 241

middle chunks are flat and empty.

The only one is a **start** chunk. No recombination of chunks can satisfy the request: generators are **material-bound**.

Annotated chunk library: VGLC overworld levels, stride 8 (12 start, 241 middle, 11 ending).

OUR APPROACH

Split the job: what was asked, and how to build it.

REQUEST

"A peaceful stroll: no enemies, no gaps, just a few coins."

→

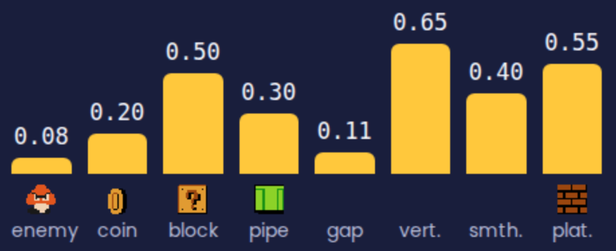
LANGUAGE BRIDGE

GPT-5.2

structured output

→

NUMERIC CONTRACT · $T \in [0,1]^8$



→

GENERATOR

MaskablePPO

compose chunks
→ gated edits
→ polish

→

RELEASE GATE

A* agent

must finish the level

✓

LANGUAGE SIDE

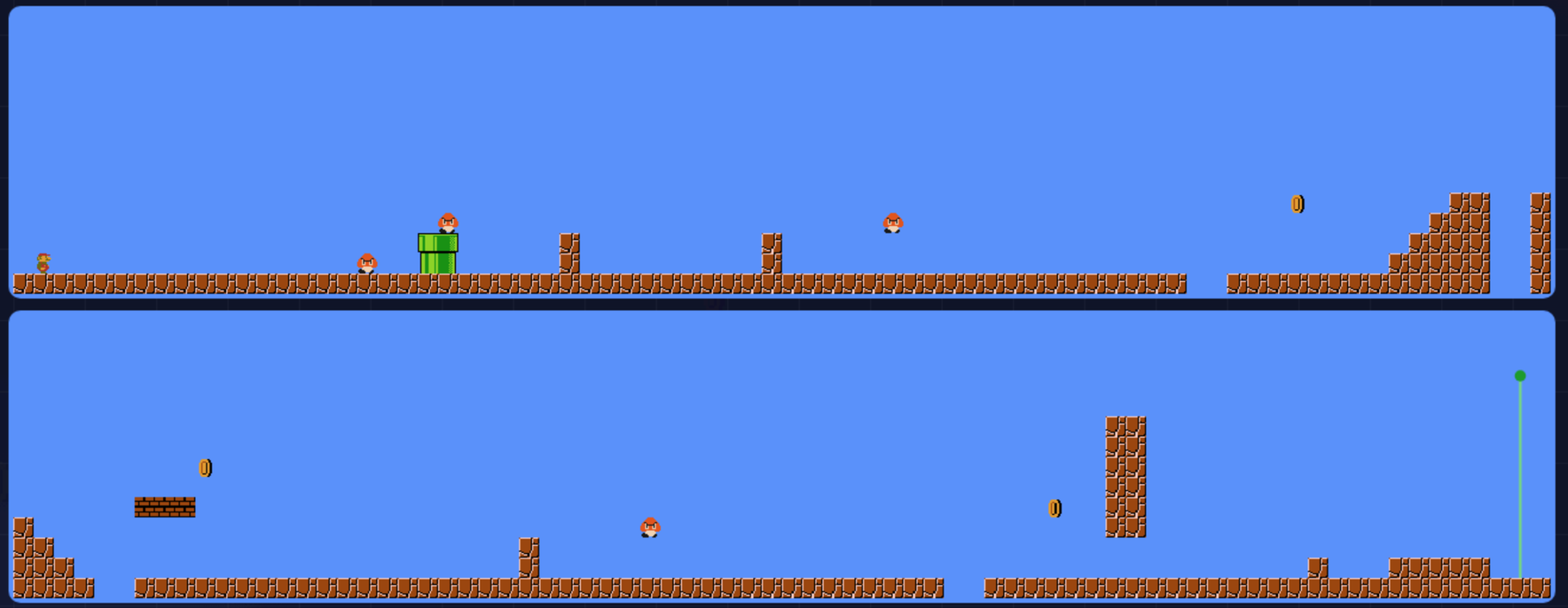
LEVEL SIDE · NO TEXT

The LLM never touches a tile.

So translation error and generation error can be measured **separately**.

INSIDE ONE EPISODE · REAL ROLLOUT

Compose from the corpus, then edit.



COMPOSE

level stitched

mean intent error

0.084

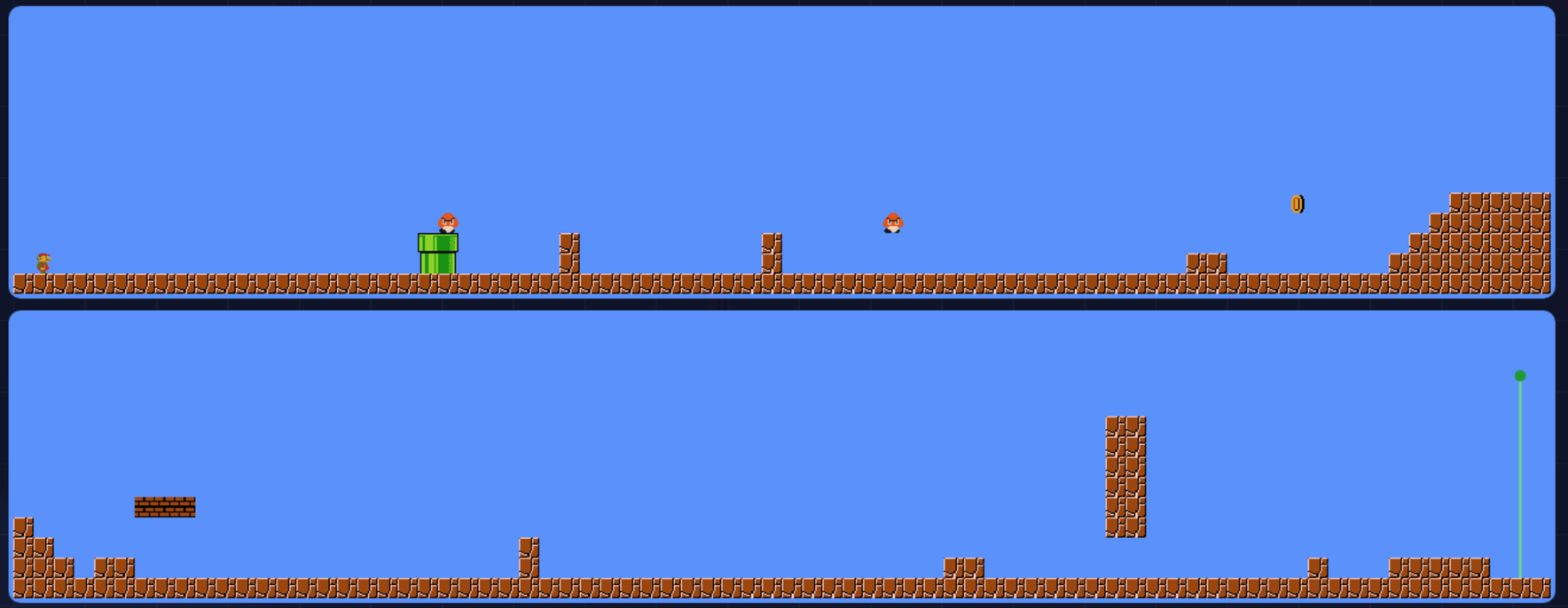
enemies	0.16
coins	0.18
blocks	0.07
pipes	0.05
gaps	0.21
verticality	0.00
smoothness	1.00
platforms	0.00

The policy appends **real corpus chunks**; masks enforce seam physics. The corpus brings enemies, gaps and coins with it.

Final 10M-step policy · target: no enemies, coins, blocks, pipes or gaps, flat, maximally smooth · every accepted decision shown.

INSIDE ONE EPISODE · REAL ROLLOUT

Compose from the corpus, then edit.



GATED EDIT

remove enemies

mean intent error

0.032

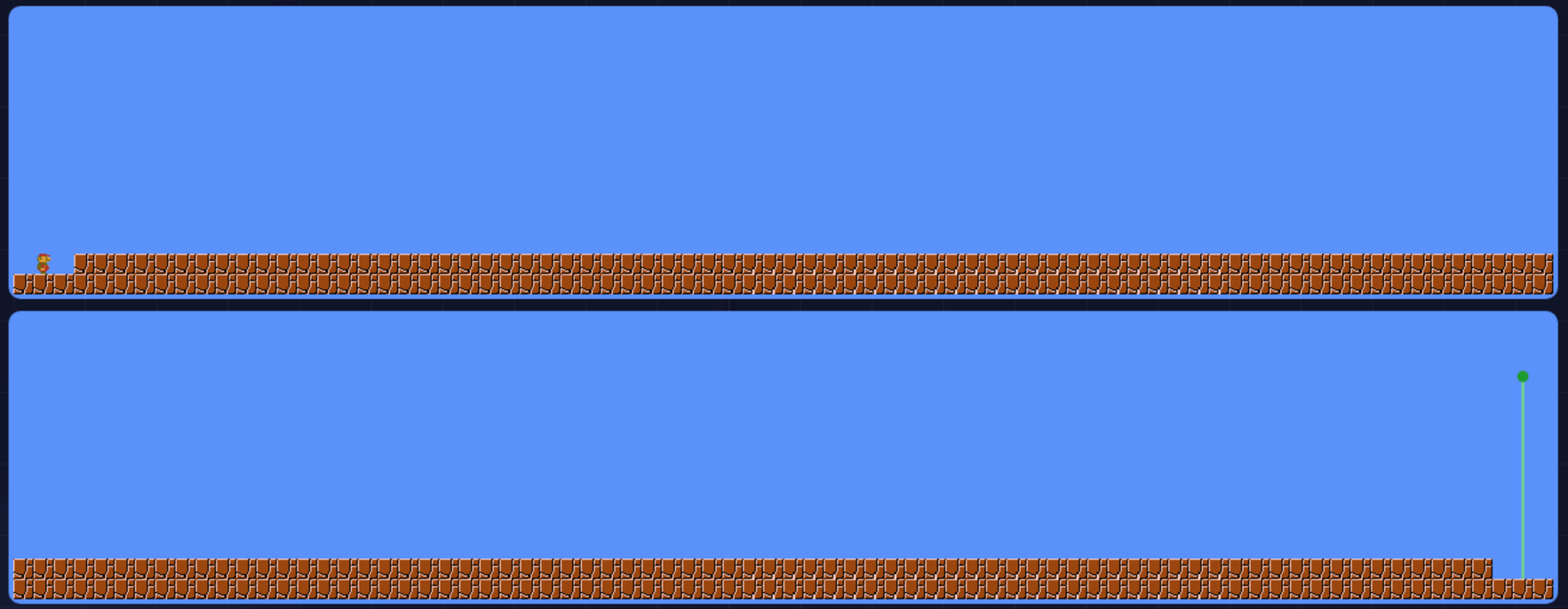
enemies		0.08
coins		0.06
blocks		0.07
pipes		0.05
gaps		0.00
verticality		0.00
smoothness		1.00
platforms		0.00

The **same policy** now picks macro-edits: remove enemies, fill gaps, strip coins. Each edit is validated and **reverted if it breaks the level**.

Final 10M-step policy · target: no enemies, coins, blocks, pipes or gaps, flat, maximally smooth · every accepted decision shown.

INSIDE ONE EPISODE · REAL ROLLOUT

Compose from the corpus, then edit.



POLISH

every dimension checked

mean intent error

0.000

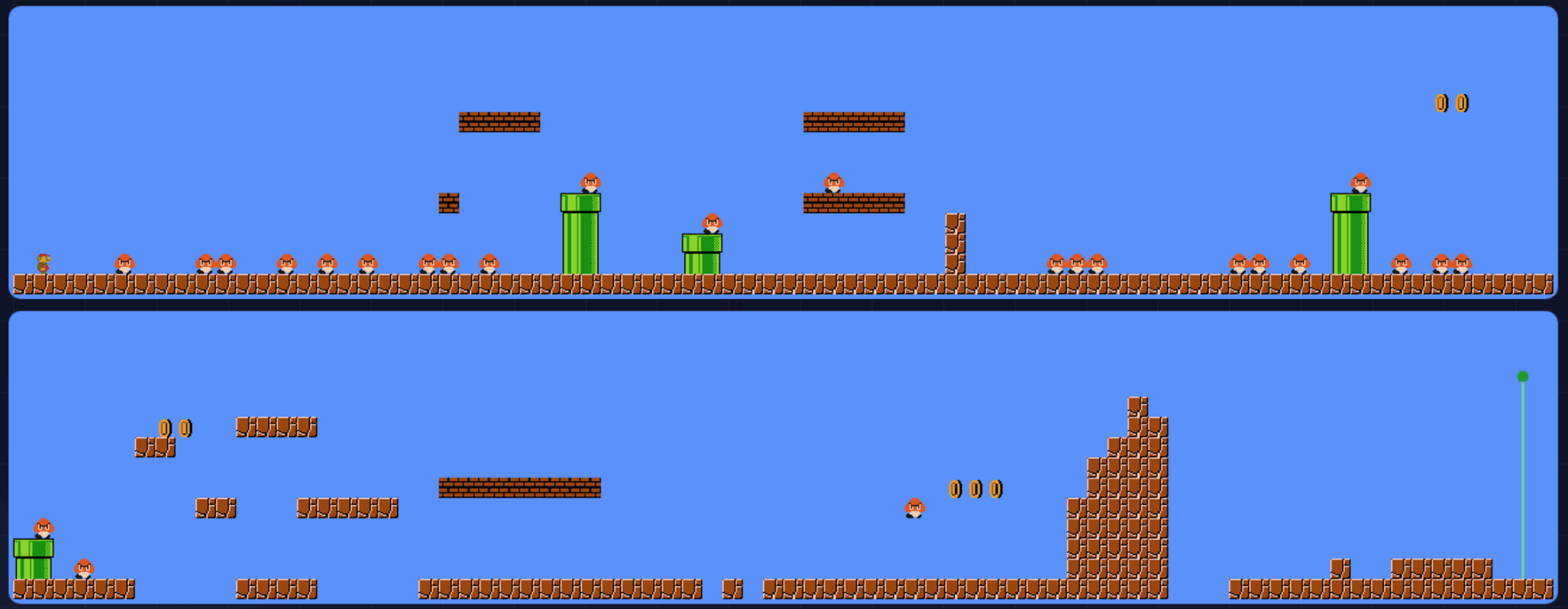
enemies	0.00
coins	0.00
blocks	0.00
pipes	0.00
gaps	0.00
verticality	0.00
smoothness	1.00
platforms	0.00

A deterministic **polish** inverts the metrics and closes the residual. One terminal reward for the whole episode, $\gamma = 1$.

Final 10M-step policy · target: no enemies, coins, blocks, pipes or gaps, flat, maximally smooth · every accepted decision shown.

CONTROLLABILITY

Zero means zero. One means as many as it can.



target enemy density

1.00

goombas in the level

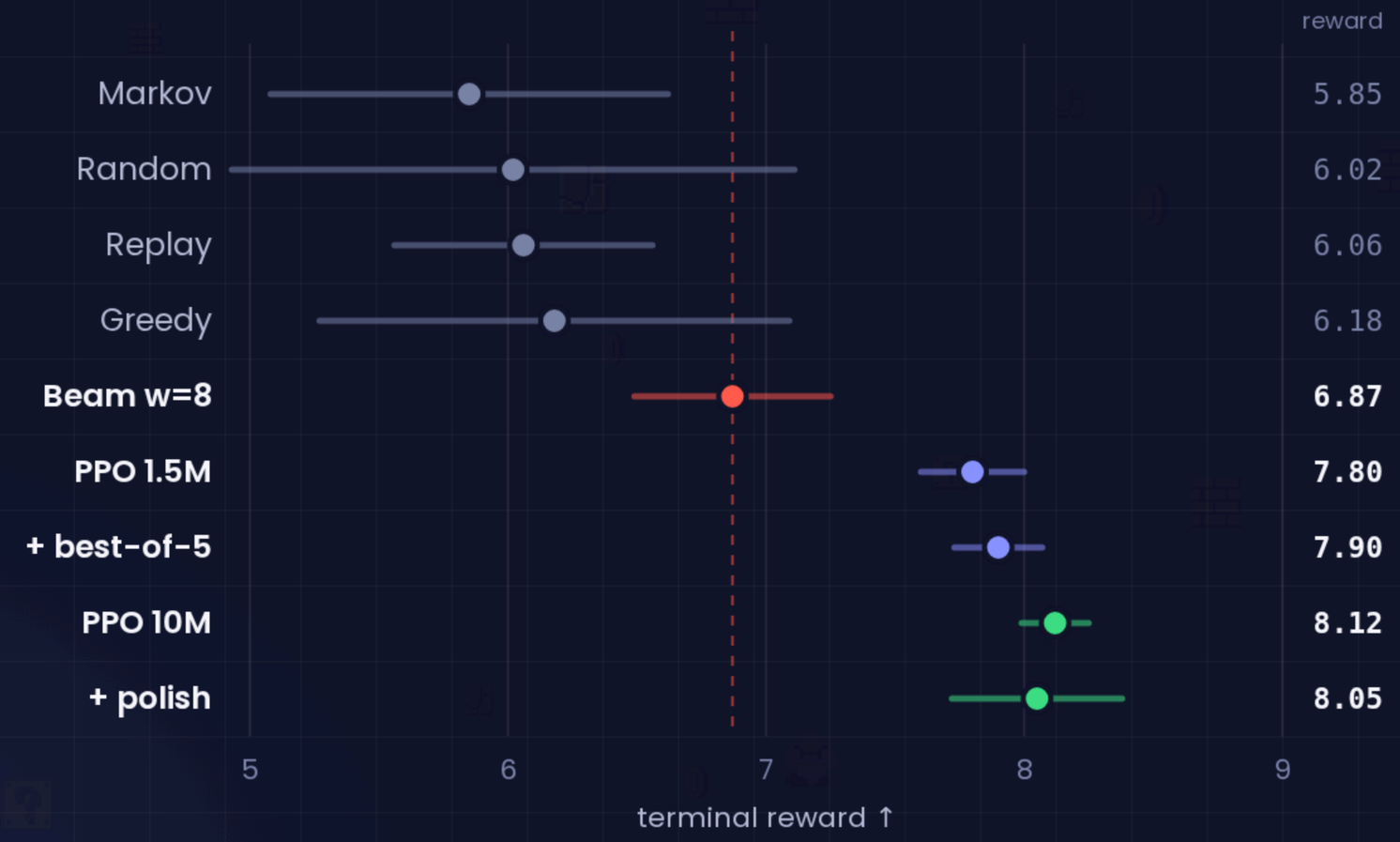
25

8/8 dimensions pass the 0→1 sweep (MAE ≤ 0.10, Spearman ρ ≥ 0.96), including exact zero, which composition alone cannot express.

Final policy + polish; other dimensions held at corpus-typical defaults; each level completed by the A* agent.

RESULTS

Learning beats search in the same action space.



+1.0

terminal reward over beam search, on all 3 training seeds ($p = 10^{-4}$)

7×

faster per level: 64 ms vs 0.47 s

36-target suite (12 intents × 3 seeds); mean ± std. Medium tier = 1.5M steps (3 seeds); final = 10M steps.

RESULTS

Editing matters when the request leaves the corpus.



−30%

intent error vs compose-only on corpus-hostile targets (p = 0.0095)

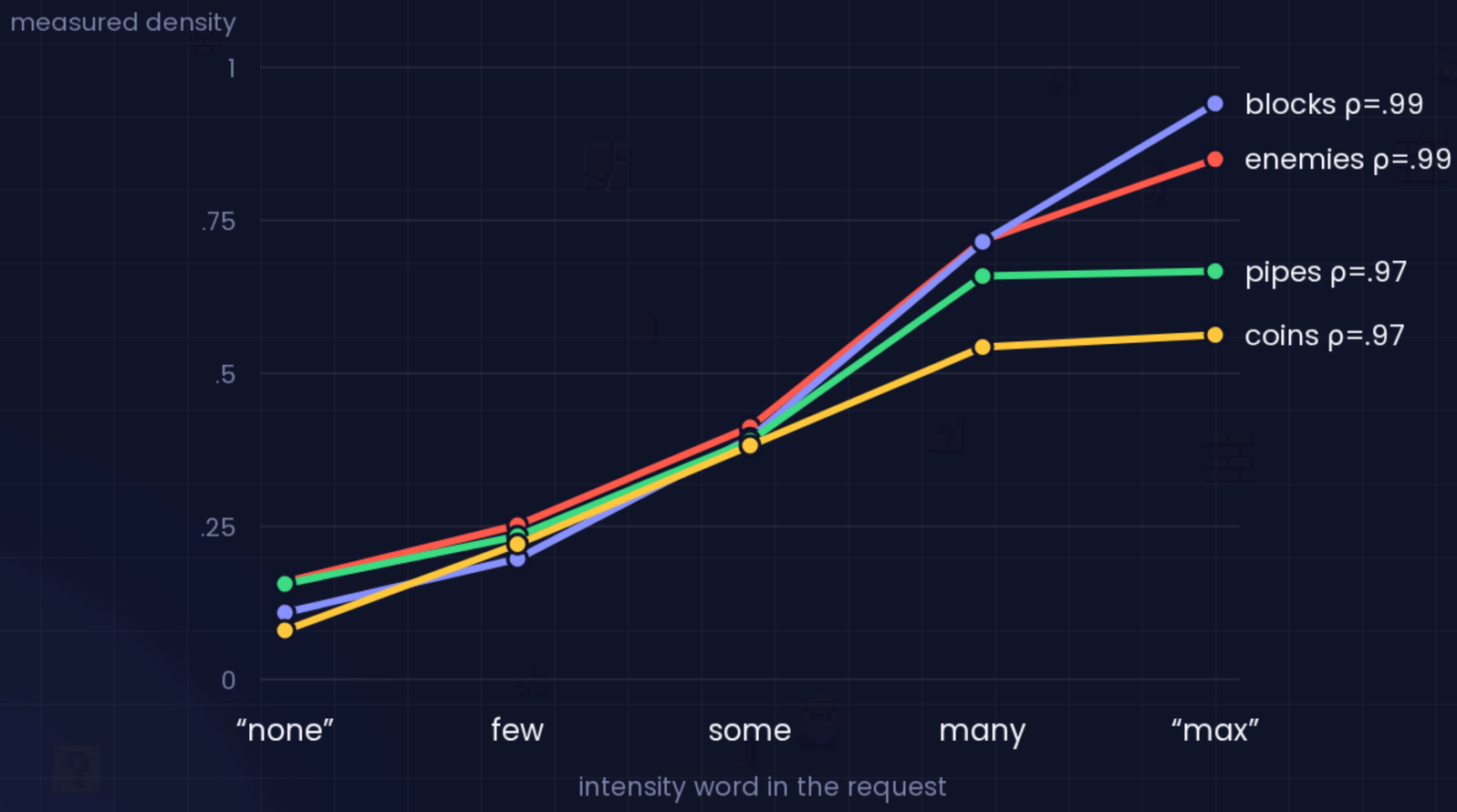
0.004

error on flat-empty after polish

Medium tier, seed 42, best-of-5. Corpus-hostile suite: six corner targets (e.g. flat-empty), n = 18.

RESULTS

Language is a reliable, monotonic interface.



0.053

translation error (MAE) over 46 descriptions

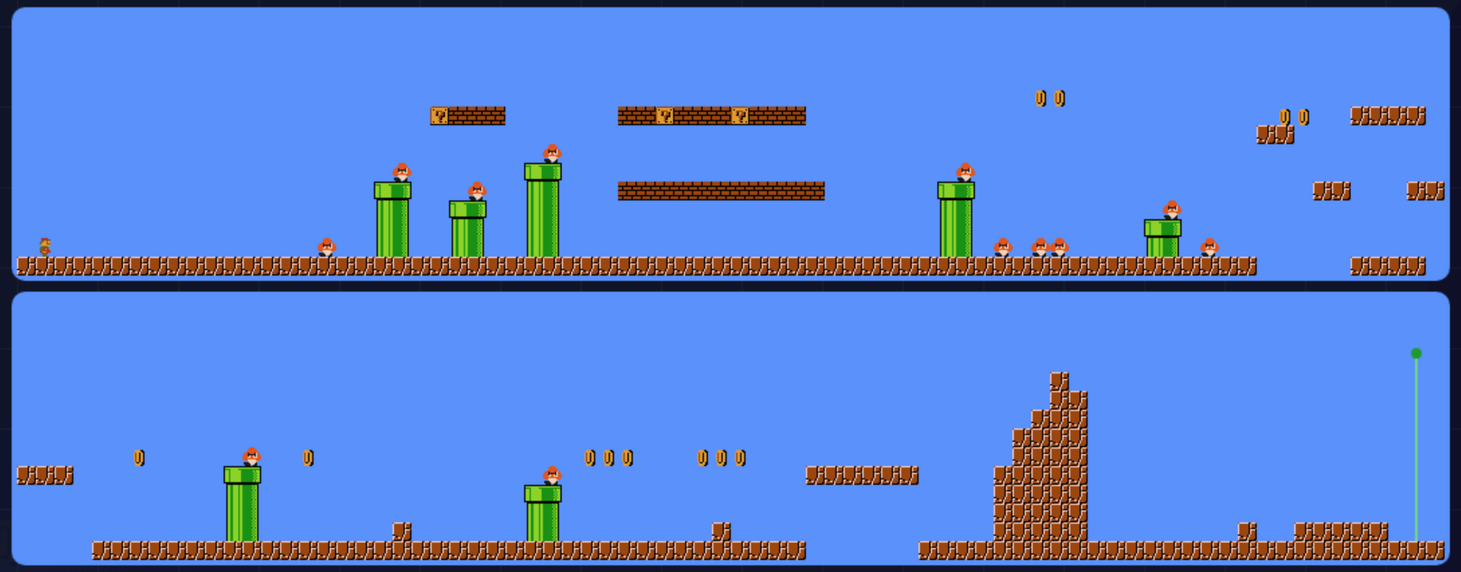
ρ ≥ .97

from “none” to “max”, end-to-end

Final model; 5 intensity bands × 3 seeds per density dimension. Translation: 46 reference-scored descriptions.

ASK, AND GET

“A garden of pipes with coins tucked between them.”



enemies	0.48	coins	0.72
blocks	0.55	pipes	0.65
gaps	0.42	verticality	0.67
smoothness	0.42	platforms	0.55



ERROR 0.025 · A* AGENT FINISHED ✓

Recorded GPT-5.2 translations, replayed exactly · final policy · ▌ = target, bar = measured.

TAKEAWAY

The request becomes a numeric contract.

- Translation missed? Inspect the vector.
- Generator missed? Per-dimension residuals.
- Targets conflict? Reported, not hidden.

Scope: structural control over a fixed 8-dimension vocabulary · one game · no designer study yet.

COME SEE THE POSTER!

slides · demo · code
github.com/brunolemosdl/intent-pcgrl